

Math 162

Problem set 4 solutions

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1. Let $F(x) := \int_a^x f(t) dt$ and $G(x) := \int_x^b f(t) dt$. We showed in class that the function F is continuous, and similarly

$$G(x) = \int_x^b f(t) dt = - \int_b^x f(t) dt$$

is continuous. Put $H := G - F$, which is a difference of continuous functions and therefore continuous.

If $\int_a^b f(t) dt = 0$, then $x = a$ and $x = b$ are solutions of the equation in question. Thus we can assume that $\int_a^b f(t) dt \neq 0$ without loss of generality. We have

$$H(a) = G(a) - F(a) = \int_a^b f(t) dt = F(b) - G(b) = -H(b),$$

which implies that either $H(a) > 0$ and $H(b) < 0$ or $H(a) < 0$ and $H(b) > 0$. Since H is continuous, the intermediate value theorem implies that there exists $x \in [a, b]$ such that $H(x) = 0$, which is precisely what we wanted to prove.

2. (a) By the power rule, we have

$$\int_1^\infty x^n dx = \lim_{N \rightarrow \infty} \int_1^N x^n dx = \lim_{N \rightarrow \infty} \frac{1}{n+1} (N^{n+1} - 1)$$

for any $n \neq -1$. If $n \geq 0$, then

$$\lim_{N \rightarrow \infty} \frac{1}{n+1} (N^{n+1} - 1) = \infty.$$

- (b) As in (a), we have

$$\int_1^\infty x^n dx = \lim_{N \rightarrow \infty} \frac{1}{n+1} (N^{n+1} - 1)$$

for any $n \neq -1$. If $n < -1$, then $n+1 < 0$ and hence

$$\lim_{N \rightarrow \infty} \frac{1}{n+1} (N^{n+1} - 1) = -\frac{1}{n+1}.$$

- (c) Following the hint, observe that

$$\int_1^{2^n} x^{-1} dx = \log(2^n) = n \log 2$$

for any $n \in \mathbb{N}$. Since $\log$ is a strictly increasing function (its derivative $\frac{1}{x}$ is positive for $x > 0$) and $2 > 1$, we have $\log 2 > 0$. Thus for any $M > 0$, if we choose $n \in \mathbb{N}$ such that $n > \frac{M}{\log 2}$, it follows that $\log(2^n) > M$.

This calculation shows that $\log N$ is an unbounded function of N on $[a, \infty)$ for any $a > 0$, from which one can deduce that the limit as $N \rightarrow \infty$ does not exist. Let us prove the slightly stronger

statement that $\lim_{x \rightarrow \infty} \log x = \infty$. Fix $M > 0$, so we must show that there exists $P > 0$ such that $x > P$ implies that $\log x > M$. By the previous paragraph, there exists $n \in \mathbb{N}$ such that $\log(2^n) > M$. Then for any $x > 2^n$, we have

$$\log x > \log(2^n) > M$$

as needed.

3. (a) First, note that if $f : [-N, N] \rightarrow \mathbb{R}$ is integrable and even, then $\int_0^N f(x) dx = \int_{-N}^0 f(x) dx$. (Proof sketch: if $P = \{x_0, \dots, x_n\}$ is a partition of $[0, N]$, then $-P := \{-x_n, \dots, -x_0\}$ is a partition of $[-N, 0]$. Using the assumption that f is even, we see that $U(f, P) = U(f, -P)$ and also that $L(f, P) = L(f, -P)$, from which the claim follows.)

Applying this to $f(x) = \frac{1}{1+x^2}$, it follows that the limit $\lim_{N \rightarrow \infty} \int_0^N \frac{1}{1+x^2} dx$ exists if and only if $\lim_{N \rightarrow \infty} \int_{-N}^0 \frac{1}{1+x^2} dx$ exists, so it suffices to consider the former. Moreover, we have

$$\int_0^N \frac{1}{1+x^2} dx = \int_0^1 \frac{1}{1+x^2} dx + \int_1^N \frac{1}{1+x^2} dx,$$

so it is enough to prove that the improper integral $\int_1^\infty \frac{1}{1+x^2} dx$ exists. Since $\frac{1}{1+x^2} < \frac{1}{x^2}$ for any $x > 0$, it follows that

$$\int_1^N \frac{1}{1+x^2} dx \leq \int_1^N \frac{1}{x^2} dx$$

for any $N \geq 1$. By 2(b) we have

$$\int_1^\infty \frac{1}{x^2} dx = 1,$$

and since $\int_1^N \frac{1}{x^2} dx$ is an increasing function of N (e.g. because its derivative is $\frac{1}{N^2} > 0$ by the fundamental theorem of calculus), it follows that $\int_1^N \frac{1}{x^2} dx \leq 1$ for any $N \geq 1$. Combining these inequalities, we obtain that $\int_1^N \frac{1}{1+x^2} dx \leq 1$ for any $N \geq 1$. Since $\int_1^N \frac{1}{1+x^2} dx$ is an increasing function of N (same argument as for $\int_1^N \frac{1}{x^2} dx$), it follows from the hint I emailed to everyone that the limit as $N \rightarrow \infty$ exists.

- (b) To say that

$$\int_{-\infty}^\infty f(x) dx = \int_0^\infty f(x) dx + \int_{-\infty}^0 f(x) dx$$

exists means both improper integrals on the right side exist. Then we have

$$\begin{aligned} \int_0^\infty f(x) dx + \int_{-\infty}^0 f(x) dx &= \lim_{N \rightarrow \infty} \int_0^N f(x) dx + \lim_{N \rightarrow \infty} \int_{-N}^0 f(x) dx \\ &= \lim_{N \rightarrow \infty} \left(\int_0^N f(x) dx + \int_{-N}^0 f(x) dx \right) \\ &= \lim_{N \rightarrow \infty} \int_{-N}^N f(x) dx, \end{aligned}$$

as desired.

- (c) For any $N \geq 0$, we have

$$\int_{-N}^N x dx = \frac{1}{2}(N^2 - (-N)^2) = 0$$

and hence

$$\lim_{N \rightarrow \infty} \int_{-N}^N x dx = 0.$$

On the other hand, since

$$\int_0^N x \, dx = \frac{1}{2}N^2,$$

we have

$$\int_0^\infty x \, dx = \lim_{N \rightarrow \infty} \frac{1}{2}N^2 = \infty.$$

Thus $\int_{-\infty}^\infty x \, dx = \int_0^\infty x \, dx + \int_{-\infty}^0 x \, dx$ does not exist.

4. (a) For any $a > 0$, we have

$$\int_a^b \frac{1}{\sqrt{x}} \, dx = 2(\sqrt{b} - \sqrt{a})$$

by the power rule. Thus

$$\int_0^b \frac{1}{\sqrt{x}} \, dx = \lim_{a \rightarrow 0^+} \int_a^b \frac{1}{\sqrt{x}} \, dx = 2\sqrt{b}.$$

(b) For any $a > 0$, we have

$$\int_a^b \frac{1}{x} \, dx = \log b - \log a.$$

Now observe that

$$\lim_{a \rightarrow 0^+} \log a = \lim_{a \rightarrow \infty} \log \frac{1}{a} = - \lim_{a \rightarrow \infty} \log a = -\infty$$

by my solution to 2(c). In particular, the improper integral

$$\int_0^b \frac{1}{x} \, dx = \lim_{a \rightarrow 0^+} (\log b - \log a)$$

does not exist.

5. We define

$$\int_{-1}^1 \frac{1}{\sqrt{1-x^2}} \, dx := \int_0^1 \frac{1}{\sqrt{1-x^2}} \, dx + \int_{-1}^0 \frac{1}{\sqrt{1-x^2}} \, dx.$$

As in 3(a), we use the fact that $\frac{1}{\sqrt{1-x^2}}$ is an even function of x to reduce to proving that $\int_{-1}^0 \frac{1}{\sqrt{1-x^2}} \, dx$ exists. Note that for $-1 < x < 0$, we have $\sqrt{1-x} > 1$ and hence

$$\frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x}} \cdot \frac{1}{\sqrt{1+x}} < \frac{1}{\sqrt{1+x}}.$$

It follows that

$$\int_a^0 \frac{1}{\sqrt{1-x^2}} \, dx \leq \int_a^0 \frac{1}{\sqrt{1+x}} \, dx$$

for any $-1 < a < 0$. We have

$$\int_a^0 \frac{1}{\sqrt{1+x}} \, dx = 2(1 - \sqrt{1+a})$$

by the power rule, which implies that

$$\int_{-1}^0 \frac{1}{\sqrt{1+x}} \, dx = \lim_{a \rightarrow -1^+} 2(1 - \sqrt{1+a}) = 2.$$

Since $\int_a^0 \frac{1}{\sqrt{1+x}} \, dx$ is a decreasing function of a (apply the fundamental theorem of calculus), it follows that

$$\int_a^0 \frac{1}{\sqrt{1+x}} \, dx \leq \int_{-1}^0 \frac{1}{\sqrt{1+x}} \, dx = 2$$

for all $-1 < a < 0$. Combining the above inequalities, we obtain

$$\int_a^0 \frac{1}{\sqrt{1-x^2}} dx \leq 2$$

for all $-1 < a < 0$. Thus $\int_a^0 \frac{1}{\sqrt{1-x^2}} dx$ is decreasing and bounded above as a function of a , so by a variant of the hint I gave via email (proved the same way, or deduced by relating right-hand limits with limits to infinity), it follows that the limit as $a \rightarrow -1^+$ exists.